

Mastercard Interview Process and Questions

Mastercard conducts a structured and competitive interview process for technical roles, which consists of four main rounds. Below is a detailed breakdown of the process, along with relevant questions and topics to prepare for each round.

1. Technical Screening Round (Initial Filter)

This round is typically conducted by a technical recruiter or engineer to evaluate the candidate's basic technical knowledge and problem-solving abilities.

Key Areas to Focus On:

- Core Java concepts
- Spring Boot fundamentals
- Problem-solving skills
- Basic coding

Questions:

- What is the difference between HashMap and ConcurrentHashMap in Java?
- What is difference in callable and runnable
- How does garbage collection work in Java?
- Explain the life cycle of a Spring Bean.
- What is Dependency Injection, and why is it important in Spring?
- How does the Spring Boot `@Transactional` annotation work?

Coding Question:

- Write a function to find the second largest number in an array.
 - Write a program to convert digits to words.
 - Implement crud service in spring boot
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2. Technical Round 1 (Super Day)

This is a deep technical discussion where the interviewer assesses your understanding of backend development, databases, and architecture.

Key Areas to Focus On:

- Advanced Java concepts
- Spring Boot and Microservices architecture
- Database interactions
- API development and REST principles
- Design patterns

Questions:

- Multiple handson questions on java streams. Using multiple functions in streams.
- What is the difference between `@RestController` and `@Controller` in Spring Boot?
- How does Spring Boot handle security by default?
- What is circuit breaker design in Microservices, and how would you implement it in Spring Boot?
- How would you optimize a query in a database with millions of rows?
- What are design patterns? Explain with example

Coding Question:

Write a program to check whether a given number is a palindrome.

3. Technical Round 2 (Super Day)

This round focuses more on system design, problem-solving, and scenario-based questions.

Key Areas to Focus On:

- System design concepts
- Microservices communication
- Performance optimization
- Transaction management in Spring Boot
- Exception handling

Sample Questions:

- How would you design a payment processing system for high scalability?
- Explain different transaction propagation levels in Spring Boot.
- How does `@ControllerAdvice` help in exception handling?

- What are the different isolation levels in Spring transactions, and when would you use each?
- Explain the concept of idempotency in APIs.
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Sample Coding Question:

Write a function to find the longest common substring between two strings.

Write a program to check parenthesis

Follow-Up:

Discuss time complexity and potential optimizations.

4. Hiring Manager Round (Super Day)

This round assesses your cultural fit, problem-solving approach, and ability to communicate technical concepts effectively.

Key Areas to Focus On:

- Behavioral questions
- Problem-solving approach
- Communication skills
- Understanding of Mastercard's business and role requirements

Questions:

- Describe a challenging project you worked on and how you resolved any issues.
- How do you prioritize tasks when working on multiple projects?
- How would you handle a situation where your team is facing continuous production issues?
- Why do you want to join Mastercard?
- What do you know about Mastercard's technological initiatives?

Technical Questions:

- Explain how you would optimize the performance of a Spring Boot application.
- What are the key considerations when designing a microservices architecture?
- How do you ensure the security of a Spring Boot application?
- Coding question - rotate string

